parts of atoms, gradually developed during the years that followed. An amazing fact was that only negative electricity could be observed in a free state, as electrons, while positive electricity would always appear in association with atoms of matter. This fact of experience indicated that atoms contained negative electrons as component parts, and therefore, free negative electricity could manifest itself only when an electron was torn off an atom, with the result that an equivalent quantity of positive electricity remained bound to the remainder of the atom. But fifty years ago it was totally impossible to arrive at a clear notion of these phenomena. The weights of atoms were approximately known, and so was the volume occupied by them. It was known also that atoms possessed electrical properties, and that they contained one or more electrons. But little or nothing at all was known about their structure, and the question as to their shape could not even be asked.

The solution of this problem was reserved for the twentieth century, the threshold of which we are now approaching in our survey of the history of atomic theory. The further course of this history is so closely linked with the topic proper of this book that in the following chapters it may be presented in connection with the latter only.